



- Constituent College of JSS Science and Technology University
- Approved by A.I.C.T.E
- Governed by the Grant-in-Aid Rules of Government of Karnataka
- Identified as lead Institution for World Bank Assistance under TEQIP Scheme

**DEPARTMENT OF PHYSICS****FIRST / SECOND SEMESTER**

| DEPARTMENT: PHYSICS                                  |                             |                                   |                                |                                       |                         |       |
|------------------------------------------------------|-----------------------------|-----------------------------------|--------------------------------|---------------------------------------|-------------------------|-------|
| Course Code: 24PH110/ 24PH210                        |                             | Course Title: ENGINEERING PHYSICS |                                |                                       |                         |       |
| Total Credits: 04<br>L-39: T-00: P-13: SL and TW- 39 |                             | SDG addressed: 4, 5, 9, 16, 17    |                                | Course Category: Basic Science Course | Course Type: Integrated |       |
| Teaching Learning Process                            |                             | Total Contact Hours/semester      | Assessment Weightage and Marks |                                       |                         |       |
|                                                      | Lecture                     | 39                                |                                | CIE                                   | SEE                     | Total |
|                                                      | Tutorial                    | 00                                | Weightage %                    | 40                                    | 60                      | 100   |
|                                                      | Practical                   | 26                                | Maximum Marks                  | 40                                    | 60                      | 100   |
|                                                      | Self-learning and Term Work | 39                                | Minimum Marks                  | 20                                    | 25                      | 45*   |
|                                                      | Total                       | 104                               |                                |                                       |                         |       |

Note: \*For passing the student must score a Minimum of 45 Marks (CIE+SEE: 20 + 25 or 21 + 24)

**Course Pre-Requisite:** Physics as one of the major subjects in Class XI and XII.

**Course Objective:**

- Educate the students in the fundamental areas of physics and its necessity in engineering and technology.
- Provide basic exposure to the principles, concepts, and techniques in physics to design proper solutions and solve practical engineering problems.
- Instigate the interest in students to work in multi-disciplinary areas of engineering, to develop an ability to analyze and adapt to the new developments in science and technology.

**Course Outcomes (COs):**

By the end of the course, students will be able to:

| <b>CO</b>  | <b>Course Outcomes</b>                                                                                                                                                                                                                                                                                                                      | <b>Highest Level of Cognitive Domain</b> | <b>Working Knowledge</b> |
|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|--------------------------|
| <b>CO1</b> | Utilize the fundamental principles of quantum mechanics to develop the solutions for few complex problems and describe the probability theory, operator formalism and tunneling.                                                                                                                                                            | L2                                       | WK1, WK3                 |
| <b>CO2</b> | Explain the basic quantum computing concepts like entanglement, superposition, qubit, gates, operators, and algorithms.                                                                                                                                                                                                                     | L2                                       | WK2, WK6                 |
| <b>CO3</b> | Elucidate the theories involved with the intrinsic and extrinsic semiconductors based on Fermi level concept to comprehend the conductivity phenomena and Hall effect. Explain the importance of dielectric properties and mechanisms in illustrating their conceptual theories and various types leading to the new engineering materials. | L2                                       | WK1, WK4                 |

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|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----------|
| <b>CO4</b> | Conceive the basic principles, Characteristics, and requisites of laser system to explain the working of lasers and their applications and use the same in explaining the propagation mechanism and its relevant concepts in the optical fiber classification and its applications. | L2 | WK7, WK8 |
| <b>CO5</b> | Demonstrate the practical skills, apply physics principles, and explain the fundamental and basic mechanisms involved in the optical, electrical and mechanical experiments to Identify the valid results and craft a well-written laboratory report.                               | L2 | WK5, WK8 |

L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate, L6 - Create

### Course Syllabus:

| Unit No. | Content                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Lecture Hours |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| 1        | <p style="text-align: center;"><b>UNIT – I: QUANTUM MECHANICS</b></p> <p>Wave function - physical significance, properties, and normalization. Time-independent Schrödinger wave equation – Derivation, Application of Schrödinger wave equation – Eigenvalue equations and Eigen functions for a particle in one-dimensional infinite potential. Expectation values and Operators (Qualitative). Barrier penetration/tunneling effect.</p> <p><b>Self-Learning and Term work:</b><br/> Solve problems involving wave functions and operators.<br/> Practice deriving and interpreting Schrödinger's equation.<br/> Start with classical physics and progress to quantum concepts.<br/> Summarize chapters in your own words.<br/> Self-assessment quizzes with solutions.<br/> Integrate math and theory through practice.</p>             | 09            |
| 2        | <p style="text-align: center;"><b>UNIT – II: QUANTUM COMPUTING</b></p> <p>Introduction to Quantum Computing. Differences between classical &amp; quantum computing. Bra and Ket notations. Quantum superposition and entanglement, the concept of qubit and its representation using Bloch sphere. Quantum gates (X, Y, Z, S and H) operation of the same on <math> 0\rangle</math> and <math> 1\rangle</math>. Quantum algorithms and Quantum circuits (Qualitative).</p> <p><b>Self-Learning and Term work:</b><br/> Create visual diagrams of quantum gates and their effects.<br/> Write analogies comparing classical bits vs. qubits.<br/> Annotate chapters with questions and insights.<br/> Solve matrix operations for quantum gates.<br/> Practice Bloch sphere visualizations.<br/> Self-assessment quizzes with solutions.</p> | 08            |

|   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |    |
|---|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 3 | <p style="text-align: center;"><b>UNIT – III: SEMICONDUCTOR PHYSICS &amp; DIELECTRICS</b></p> <p>Direct and Indirect band gap semiconductors. Electron and hole concentrations at equilibrium in intrinsic semiconductors (derivation for electron only). Fermi level in an intrinsic semiconductor. Intrinsic density and conductivity (No derivation). Extrinsic conductivity – Temperature dependence of carrier concentration (Qualitative). Fermi level in extrinsic semiconductors. Hall Effect in semiconductors.</p> <p>Dielectric Polarization and types. Internal fields in dielectrics. Clausius – Mosotti equation (No derivation). Dielectric constant. Frequency dependence of dielectric constant. Dielectric loss concept. Electrostriction. Ferroelectrics – principle and hysteresis theory. Piezoelectric – principle and mechanism.</p> <p><b>Self-Learning and Term work:</b><br/> Elaborate on the importance of semiconductor knowledge in advancement of technology. Demonstrate the use of Hall effect in determination of the type and concentration of charge carriers.<br/> Explain the doping concentration effect on electrical properties of the semiconductor.<br/> Write an essay on the type of polarization mechanisms in dielectric.<br/> Explore the applications of dielectrics in storage devices.</p> | 12 |
| 4 | <p style="text-align: center;"><b>UNIT – IV: LASERS &amp; OPTICAL FIBERS</b></p> <p>Interaction of radiation with matter. Characteristics of Lasers. Requisites &amp; conditions for a Laser system. He – Ne Laser. Applications – Laser printing.<br/> Propagation mechanism in optical fibers – TIR. Angle of acceptance &amp; Numerical aperture – Derivation and propagation condition. Types of optical fibers. Attenuation (Qualitative) &amp; Attenuation coefficient (No derivation). Application – Point to point communication systems.</p> <p><b>Self-Learning &amp; Term work:</b><br/> Watch animations or simulations of laser operation and summarize the process.<br/> Research one of the laser applications and present its working principle.<br/> Write a short essay on how lasers have transformed one field.<br/> Use a laser pointer and any substance to show TIR.<br/> Explore attenuation using graphs and real-world data.<br/> Write a short essay on how optical fibers revolutionized internet speed.</p>                                                                                                                                                                                                                                                                                                      | 10 |
| 5 | <p style="text-align: center;"><b>UNIT – V: ENGINEERING PHYSICS LABORATORY</b></p> <p>Diffraction Grating - Determination of wavelength of spectral lines of mercury source of light using a diffraction grating.<br/> Newton's Rings - Determination of radius of curvature of a plano-convex lens by forming Newton's rings using a monochromatic source of light (sodium).<br/> Uniform Bending - Determination of Young's modulus of the material of the bar by uniform bending.<br/> Torsional Pendulum - Determination of moment of inertia of an irregular body, using objects of regular shape and to determine the rigidity modulus of the material of the wire.<br/> Four Probe Method – Determination of the energy gap and the resistivity of an n-type germanium semiconductor sample by using a four-probe setup.<br/> Dielectric Constant - Determination of the dielectric constant of the dielectric material used in a capacitor by charging and discharging of the capacitor.<br/> Fermi Energy - Determination of the Fermi energy and Fermi temperature of the given</p>                                                                                                                                                                                                                                                 | 12 |

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|  | <p>material copper wire.</p> <p>Planck's constant – Determination of Planck's constant via the current – voltage characteristics study of LEDs.</p> <p>Laser Characteristics – Determination of spot size and divergence angle of a laser beam via the study of Gaussian and Polarization profile of laser beam.</p> <p>0. Optical Fiber Characteristics – Determination of band width of given optical Fiber cable (OFC) by studying the intensity profile of Laser.</p> <p>1. Hall Effect – Determination of Magnetic sensitivity, Hall voltage, Hall coefficient, charge concentration and velocity of charge inside the crystal lattice of p-type semiconductor.</p> <p><b>Self-Learning and Term work:</b></p> <p>Understand the concept of least count, accuracy, the relation between the same.</p> <p>Practice problems on circuit and non-circuit experiments.</p> <p>Each group is given different types of problems (forward interpolation, backward interpolation, Lagrange) and later present solutions.</p> |  |
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#### Text Books:

| Sl. No | Author/s                                         | Title                                       | Publisher Details                                                |
|--------|--------------------------------------------------|---------------------------------------------|------------------------------------------------------------------|
| 1      | Gaur R K & Gupta S L                             | Engineering Physics                         | Dhanpathrai & Sons, 3 <sup>rd</sup> Edition, 2016                |
| 2      | Avadhanulu M N, Kshirsagar P G & Arun Murthy TVS | Engineering Physics                         | S Chand & Publications Pvt. Ltd., 11 <sup>th</sup> Edition, 2018 |
| 3      | Michael A. Nielsen & Isaac L. Chuang             | Quantum Computation and Quantum Information | Cambridge University Press, 12 <sup>th</sup> Edition, 2013       |
| 4      | Malik H K and Singh A K                          | Engineering Physics                         | Tata Mc Graw Hill, 2 <sup>nd</sup> Edition, 2017                 |

#### Reference Books:

| Sl.No. | Author/s                           | Title                                                    | Publisher Details                                                     |
|--------|------------------------------------|----------------------------------------------------------|-----------------------------------------------------------------------|
| 1      | Raymond A. Serway & John W. Jewett | Physics for Scientists and Engineers with Modern Physics | Cengage Learning India Pvt. Ltd., 10 <sup>th</sup> Edition, 2023      |
| 2      | Pillai S O                         | Solid State Physics                                      | New Age International Publishers, 10 <sup>th</sup> Edition, 2022      |
| 3      | Parag K Lala                       | Quantum Computing                                        | McGraw Hill Education Pvt. Ltd., 1 <sup>st</sup> Indian Edition, 2020 |
| 4      | Ajoy Ghatak & Thyagarajan K        | Lasers                                                   | Trinity Press, 2 <sup>nd</sup> Edition, 2019                          |
| 5      | Donald A Neamen                    | Semiconductor Physics & Devices                          | McGraw Hill, 4 <sup>th</sup> Edition, 2017                            |

#### Web/Digital resources:

| Sl.No. | Web Link                                                                                  |
|--------|-------------------------------------------------------------------------------------------|
| 1      | <a href="http://hyperphysics.phy-astr.gsu.edu/">http://hyperphysics.phy-astr.gsu.edu/</a> |
| 2      | <a href="https://atalacademy.aicte-india.org">https://atalacademy.aicte-india.org</a>     |
| 3      | <a href="https://swayam.gov.in/">https://swayam.gov.in/</a>                               |
| 4      | SJCE, JSSSTU Digital library                                                              |

**Course Articulation Matrix:**

| <b>POs →<br/>COs ↓</b> | <b>PO1</b> | <b>PO2</b> | <b>PO3</b> | <b>PO4</b> | <b>PO5</b> | <b>PO6</b> | <b>PO7</b> | <b>PO8</b> | <b>PO9</b> | <b>PO10</b> | <b>PO11</b> |
|------------------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|-------------|-------------|
| <b>CO1</b>             | 3          | 3          |            |            |            |            |            |            |            |             | 1           |
| <b>CO2</b>             | 3          | 2          |            |            |            |            |            |            |            |             | 1           |
| <b>CO3</b>             | 3          | 3          |            |            |            |            |            |            |            |             | 1           |
| <b>CO4</b>             | 3          | 2          |            |            |            |            |            |            |            |             | 1           |
| <b>CO5</b>             | 3          | 3          |            |            |            | 1          |            |            | 1          | 1           | 1           |

**High – 3, Medium – 2, Low – 1**